Class 9

MATTER IN OUR SURROUNDINGS

Ch 1 | Class 9th (Science)

Handwritten Notes

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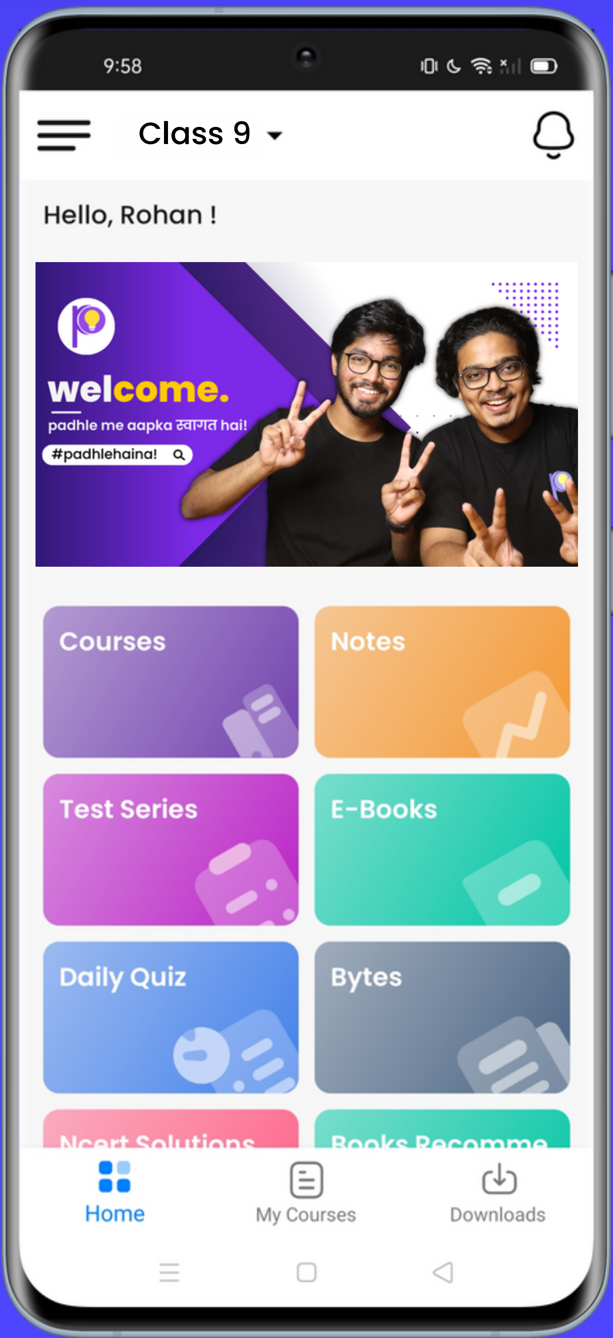

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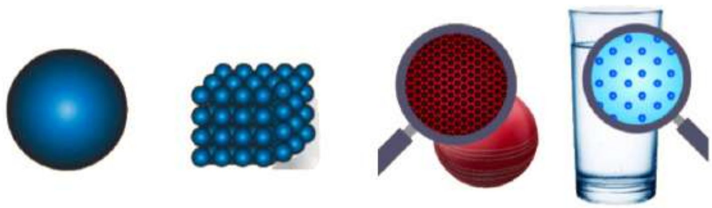
Matter In Our Surrounding

* What is Matter?

→ Matter is made up of small particles.

→ We can either see it, touch it, feel it or taste it, or all.



→ For a long time, two schools of thought prevailed regarding the nature of matter.

- One school believed matter to be continuous like a block of wood, whereas, the other thought that matter was made up of particles like sand.

* What are characteristics of particles matter

1. Particles of Matter have spaces between them.
2. <u>Particles of Matter are continuously moving</u>:

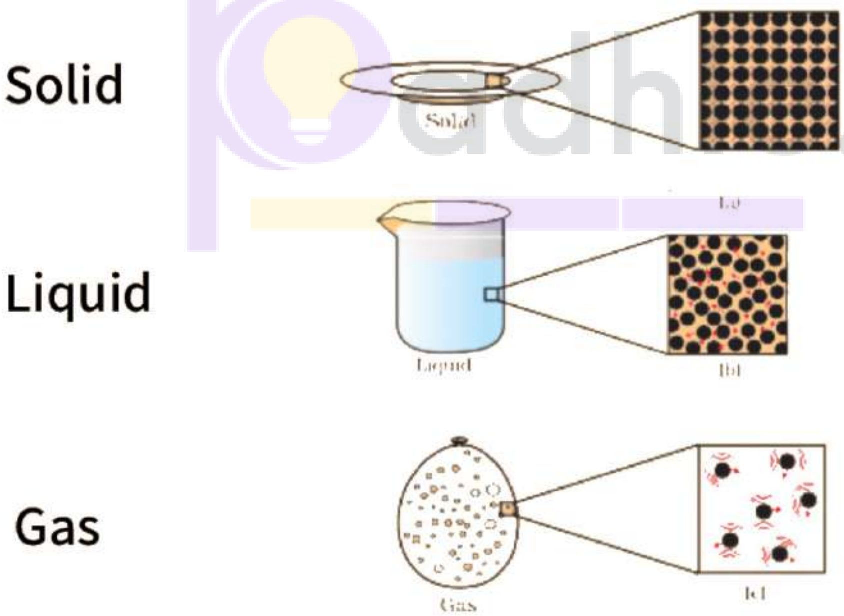
- Particles of matter are continuously moving, that is, they possess what we call the kinetic energy.
- As the temperature rises, particles move faster.
- So, we can say that with increase in temperature the kinetic energy of the particles also increases.

- Particles of matter intermix on their own with each other.
- They do so by getting into the spaces between the particles.
- This intermixing of particles of two different types of matter on their own is called difusion.

3. Particles of Matter attract eachother:

- This force keeps the particles together.
- The strength of this force of attraction varies from one kind of matter to another.

* What are States of Matter?



	Shape	Volume
Solid	Definite	Definite
Liquid	Indefinite	Indefinite
Gas	Indefinite	Indefinite

	Intermolecular Force of Attraction	Intermolecular Spaces
Solid	Maximum	Minimum
Liquid	Moderate	Moderate
Gas	Minimum	Maximum

Solid :

- Have a definite shape, distinct boundaries and fixed volumes, that is, have negligible compressibility.
- Solids have a tendency to maintain their shape when subjected to outside force.
- Solids may break under force but it is difficult to change their shape, so they are rigid.

(a) What about a rubber band, can it change its shape on stretching? Is it a solid?

(b) What about sugar and salt? When kept in different jars these take the shape of the jar. Are they solid?

(c) What about a sponge? It is a solid yet we are able to compress it. Why?

Liquids :

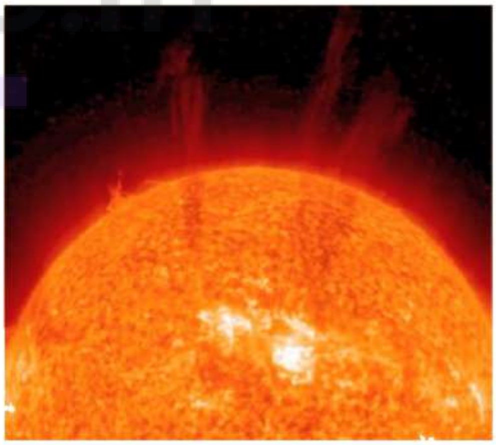
- Have a definite shape, distinct boundaries and fixed volume, that is, have negligible compressibility.
- Solids have a tendency to maintain their shape when subject to outside force.
- Solids may break under force but it is difficult to change their shape, so they are rigid.
- They take up the shape of the container in which they are kept.
    - Liquids flow and change shape, so they are not rigid but can be called fluid.
- The gases from the atmosphere diffuse and dissolve in water.
- These gases, especially oxygen and carbon dioxide, are essential for the survival of acquatic animals and plants.
- The rate of diffusion of liquids is higher than that of solids.
- This is due to the fact that in the liquid state, particles move freely and have greater space between each other as compared to particles in the solid state.

Gases :

- They take up the shape of the container in which they are kept.
  - Liquids flow and change shape, so they are not rigid but can be called fluid.
- The gases from the atmosphere diffuse and dissolve in water.
- These gases, especially oxygen and carbon dioxide, are essential for the survival of acquatic animals and plants.

* Plasma

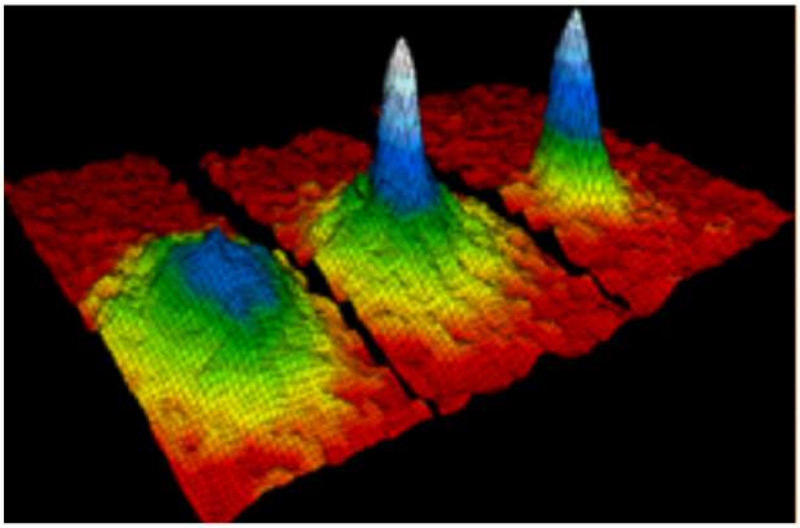
- Plasma is superheated matter - so hot that the electrons are ripped away from the atoms forming an ionised gas.
  - It comprises over 99% of the visible universe.



- Plasma is often called "the fourth state of matter", along with solid, liquid and gas.

* BEC [Bose Einstein Condensate]

- In condensed matter physics, a Bose-Einstein condensate is a state of matter which is typically formed when a gas of bosons at low densities is cooled to temperatures very close to absolute zero.



* What is Phase of Transformation?

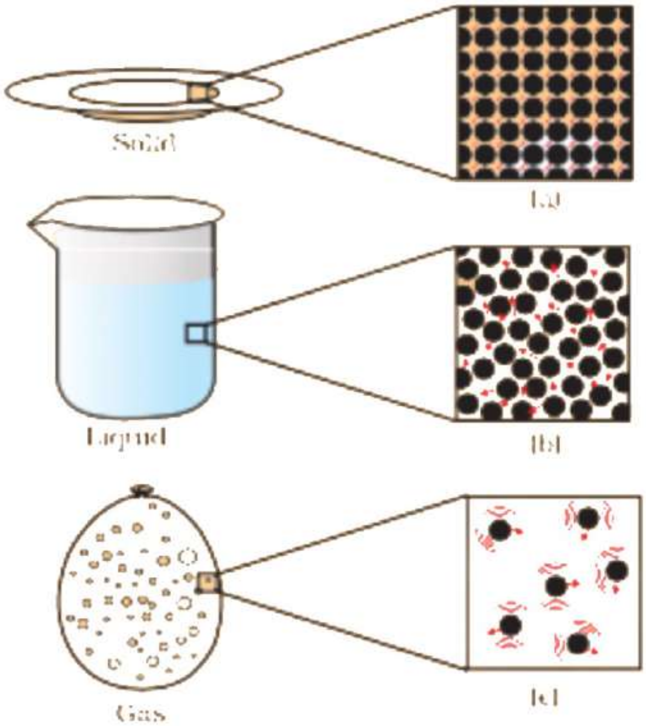
→ <u>Conversion of States</u> :

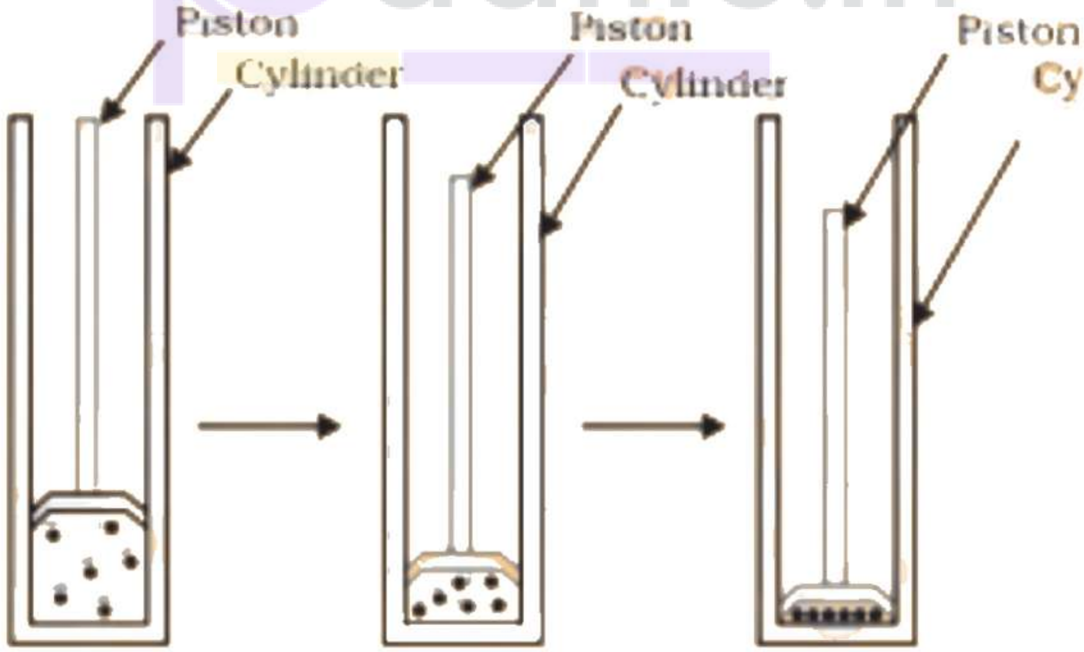
$$\boxed{\text{Solid}} \underset{\text{Cool}}{\overset{\text{Heat}}{\rightleftharpoons}} \boxed{\text{Liquid}} \underset{\text{Cool}}{\overset{\text{Heat}}{\rightleftharpoons}} \boxed{\text{Gaseous}}$$

State

$$\begin{tikzcd}
 & \boxed{\text{Liquid}} \arrow[dl, shift left] \arrow[dr, shift left] & \\
\boxed{\text{Solid}} \arrow[ur, shift left] \arrow[rr, shift left] & & \boxed{\text{Gas}} \arrow[ul, shift left] \arrow[ll, shift left]
\end{tikzcd}$$

Gas to Liquid using pressure





* Phase Transformation

- On increasing the temperature of solids, the kinetic energy of the particles increases.
- Due to the increase in kinetic energy, the particles start vibrating with greater speed.
- The energy supplied by heat overcomes the forces of attraction between the particles.
- The particles leave their fixed positions and start moving more freely.
- A stage is reached when the solid melts and is converted to a liquid.
- The minimum temperature at which solid melts to become a liquid at the atmospheric pressure is called its melting point.

- The melting point of a solid is an indication of the strength of the force of attraction between its particles.

* What is Melting Point?

- The temperature at which a substance will melt.

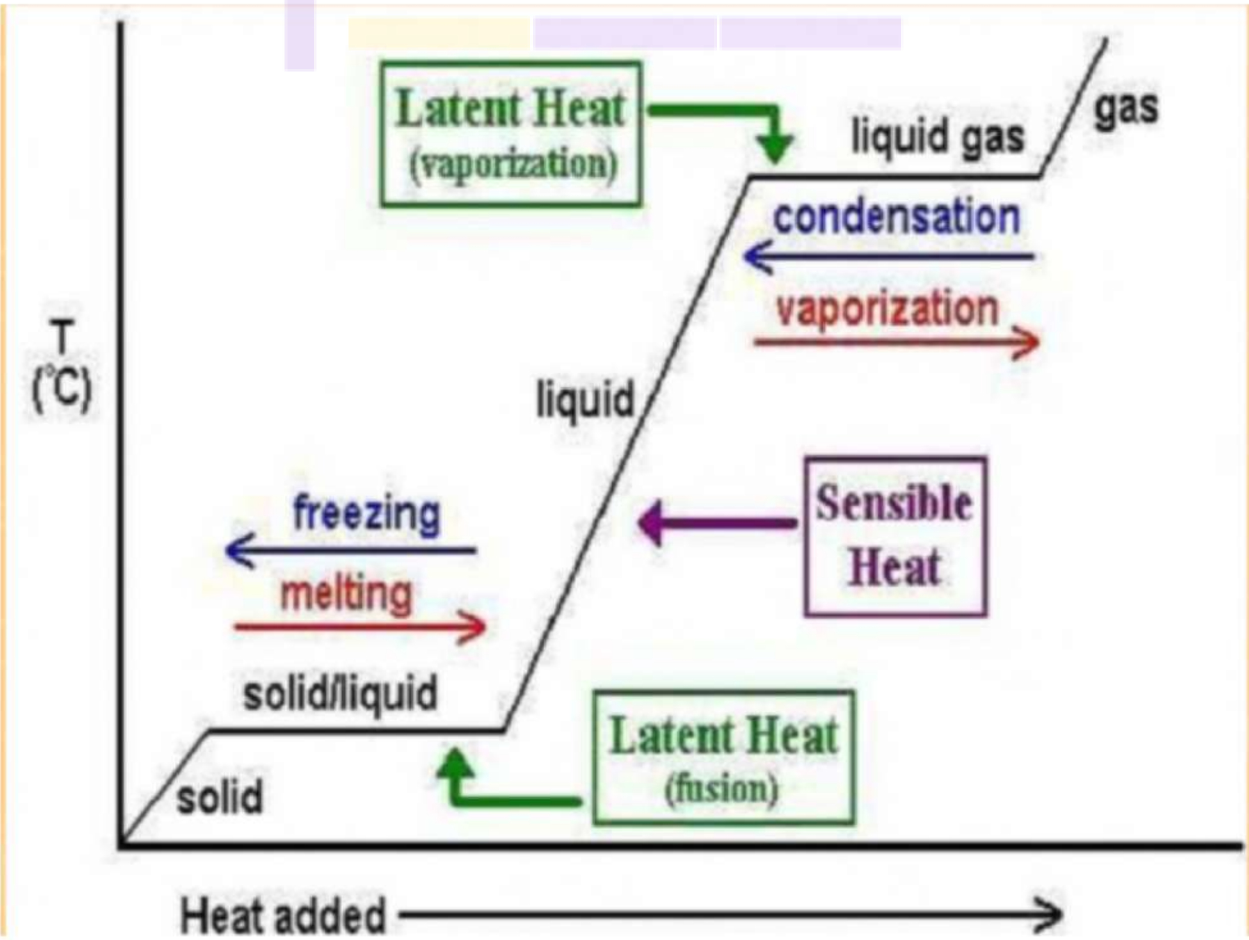
* What is Boiling Point ?

- The temperature at which a liquid starts to boil.

- During a phase change, the temperature remains constant.

- When a solid melts, its temperature remains the same, so where does the energy go?

* What is Latent Heat ?



* Latent Heat of Fusion :

- Amount of heat required to convert 1 kg of given solid into liquid.

* Latent Heat of Vaporisation :

- Amount of heat required to convert 1kg of given liquid.

Unit - joule

* What is Evaporation?

- Evaporation is the process by which a liquid turns into a gas.
- It causes cooling.

Evaporation	Boiling
Takes place only on the surface.	Takes place throughout the liquid.
Liquid temperature drops during evaporation.	Liquid temperature remains unchanged during boiling.
Evaporation is a gradual process.	Boiling is a vigorous process, with the formation of bubbles throughout the liquid.

* Factors affecting Evaporation :

1. Temperature :
- Higher temperature of liquid, higher rate as more molecules at the surface are energetic enough to escape.

2. Humidity :
- Higher humidity, lower the rate as more water vapour present in the air to hit the escaped molecule back to liquid.

3. Boiling Point :
- Lower boiling point of liquid, faster rate as bonds between liquid molecules are weaker.

4. Surface Area of Liquid :
- Larger surface area, more molecules can escape from liquid at greater rate.

5. Movement of Air :
- Removes the liquid molecules as soon as they escape.

6. Pressure :
- Reducing atmospheric pressure increases rate of evaporation.

* What are Temperature Scales ?

°C to °K

$$T_k = T_c + 273$$

°C to °F

$$°F = \frac{9}{5}°C + 32$$